

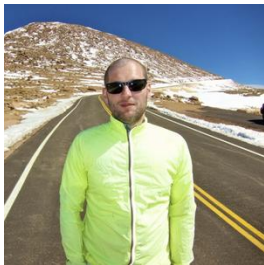
Off with the head!

Termination provers and the word problem for
1-relation monoids

Reinis Cirpons

University of St Andrews

Joint work with



James Mitchell
University of St Andrews



Finn Smith
University of St Andrews

String rewriting systems

Notation

A	Alphabet	$A = \{a, b, c\}$
A^*	Words over A	$A^* = \{\varepsilon, a, aa, ab, \dots\}$
R	String rewriting system	$R = \left\{ \begin{array}{l} ab \rightarrow ba \\ abc \rightarrow \varepsilon \end{array} \right\}$
$\rightarrow_R$	Single-step derivation	$abbc \rightarrow_R babc$
$\rightarrow_R^*$	Multi-step derivation	$abbc \rightarrow_R^* b$

The most important problem for SRS

SRS Termination problem

Given a string rewriting system R , determine if R is terminating.

Theorem (Folklore)

The SRS termination problem is undecidable.

The most important problem for SRS

n -rule SRS Termination problem

Given a rewriting system R with **at most n rules**, determine if R is terminating.

Open problem

For which n is the n -rule SRS termination problem decidable?

n	1	2	3	4	5	6	...
n -rule SRS termination							

n	1	2	3	4	5	6	...
n -rule SRS termination	?	?	X	X	X	X	

Theorem (Matiyasevich and Senizergues 1996)

The 3-rule SRS termination problem is undecidable.

n	1	2	3	4	5	6	...
n -rule SRS termination	?	?	X	X	X	X	
n -rule TRS termination	X	X	X	X	X	X	

Theorem (Matiyasevich and Senizergues 1996)

The 3-rule SRS termination problem is undecidable.

Theorem (Dauchet 1988)

The 1-rule TRS termination problem is undecidable.

Thue systems

A *Thue system* is a bidirectional SRS, e.g. $R = \left\{ \begin{array}{l} ab \leftrightarrow ba \\ abc \leftrightarrow \varepsilon \end{array} \right\}$.

We say that w_1 *equals* w_2 *modulo* R if $w_1 \leftrightarrow_R^* w_2$.

$$bbac \leftrightarrow_R^* abcb \quad \text{since} \quad bbac \leftrightarrow_R babc \leftrightarrow_R b \leftrightarrow_R abcb.$$

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Word problem

Given a Thue system R and a pair of words w_1, w_2 , determine if

$$w_1 \leftrightarrow_R^* w_2.$$

Theorem (Markov 1947 and Post 1947)

The word problem is undecidable.

Thue systems

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n -relation word problem

Given a Thue system R with **at most n relations** and a pair of words w_1, w_2 , determine if

$$w_1 \leftrightarrow_R^* w_2.$$

Open problem

For which n is the n -relation word problem decidable?

n	1	2	3	4	5	6	...
n -rule SRS termination	?	?	X	X	X	X	
n -rule SRS termination	X	X	X	X	X	X	
n -relation word problem	?	?	X	X	X	X	

Theorem (Matiyasevich 1967)

The 3-relation word problem is undecidable.

What does this have to do with monoids?

A *monoid* is a set M together with a binary operation $\cdot : M \times M \rightarrow M$ and distinguished element $1_M \in M$ such that for all $x, y, z \in M$

$$(x \cdot y) \cdot z = x \cdot (y \cdot z) \quad \text{and} \quad 1_M \cdot x = x = x \cdot 1_M.$$

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The $\leftrightarrow_R^*$ equivalence classes of a Thue system R form a monoid with multiplication given by

$$w_1 / \leftrightarrow_R^* \cdot w_2 / \leftrightarrow_R^* = (w_1 w_2) / \leftrightarrow_R^*.$$

We call this monoid the *monoid presented by* $P = \langle A \mid R \rangle$. In this talk we will refer to it as the monoid *arising from* R .

n	1	2	3	4	5	6	...
n -rule SRS termination	?	?	X	X	X	X	
n -rule TRS termination	X	X	X	X	X	X	
n -relation word problem	?	?	X	X	X	X	

n	1	2	3	4	5	6	...
n -rule SRS termination	?	?	X	X	X	X	
n -rule TRS termination	X	X	X	X	X	X	
n -relation monoid WP	?	?	X	X	X	X	

n	1	2	3	4	5	6	...
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n -rule TRS termination	X	X	X	X	X	X	
n -relation monoid WP	?	?	X	X	X	X	
n -relation group WP							
n -relation inverse monoid WP							

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n -relation monoid WP	?	?	X	X	X	X	
n -relation group WP	✓						
n -relation inverse monoid WP							

Theorem (Magnus 1932)

The 1-relation group word problem is decidable.

n	1	2	3	...	11	12	...
n -rule SRS termination	?	?	X	...	X	X	
n -rule TRS termination	X	X	X	...	X	X	
n -relation monoid WP	?	?	X	...	X	X	
n -relation group WP	✓	?	?	...	?	X	
n -relation inverse monoid WP							

Theorem (Magnus 1932)

The 1-relation group word problem is decidable.

Theorem (Borisov 1969)

The 12-relation group word problem is undecidable.

n	1	2	3	...	11	12	...
n -rule SRS termination	?	?	X	...	X	X	
n -rule TRS termination	X	X	X	...	X	X	
n -relation monoid WP	?	?	X	...	X	X	
n -relation group WP	✓	?	?	...	?	X	
n -relation inverse monoid WP	X	X	X	...	X	X	

Theorem (Magnus 1932)

The 1-relation group word problem is decidable.

Theorem (Borisov 1969)

The 12-relation group word problem is undecidable.

Theorem (Gray 2019)

The 1-relation inverse monoid word problem is undecidable.

The word problem for 1-relation monoids

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- ▶ (Oganesian 1978) If the initial letters of u and v differ and u and v have the same number of a 's in them, then the word problem is decidable, e.g. $\{baaba \leftrightarrow aabbbba\}$;

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- ▶ Many more conditions, see e.g. Nyberg-Brodda 2021.

Our goal

Question

What is the smallest instance of a 1-relation monoid presentation where the decidability of its word problem is unknown?

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What is the smallest instance of a 1-relation monoid presentation where the decidability of its word problem is unknown?

Theorem (Adian and Oganesian 1987)

The 1-relation monoid word problem reduces to the word problem for Thue systems of the form

$$\{bua \leftrightarrow ava\} \quad \text{or} \quad \{bua \leftrightarrow a\}$$

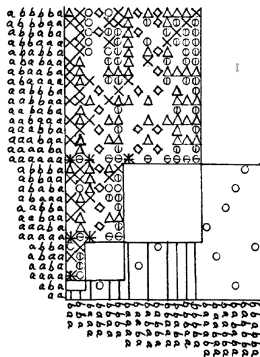
where $u, v \in \{a, b\}^*$.

We call such monoids *left-cycle free*.

Prior art Pedersen 1989, $\max(|u|, |v|) \leq 6$:

TABLE 1

Cases of $aAa = bBa$

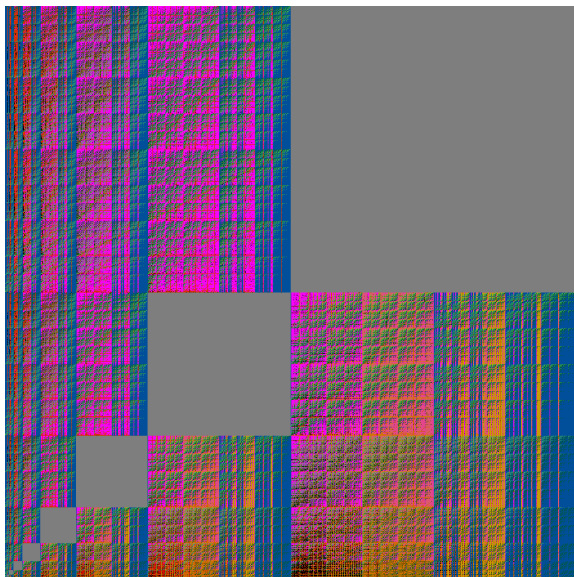


Legend

solvable by

- | left side (bBa) is hypersimple
- * [Og79]
- × [AO87]
- ⊖ Thm 1
- ⊕ Thm 2
- other cases of (S1)
- △ Thm 3 (S2)
- ◇ S3

Our results, $\max(|u|, |v|) \leq 8$:



Method	Count
Known mathematical results	222786
Knuth-Bendix	37624
Knuth-Bendix backtrack	502
Unsolved	720
Total	261632

The hard cases

Theorem (C., Mitchell, Smith 2025+)

Let R be a left-cycle free one relation Thue system. Then there exists an orthogonal SRS $\mathcal{A}_R$ such that

$\mathcal{A}_R$ is terminating $\implies R$ has decidable word problem.

Let $R = \{bua \leftrightarrow ava\}$ for some $u, v \in \{a, b\}^*$. Adian 1960a shows that left-cycle free implies left cancellative, i.e.

$$px \leftrightarrow_R^* py \iff x \leftrightarrow_R^* y.$$

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$$w_1 = paw'_1 \xleftarrow{*} w_2 = pbw'_2$$

$$aw'_1 \xleftarrow{*} bw'_2$$

$$aw'_1 \overset{*}{\longleftrightarrow} as_1 \longleftrightarrow bs_2 \overset{*}{\longleftrightarrow} bw'_2$$

$$aw'_1 \xleftrightarrow{*} avas' \longleftrightarrow buas' \xleftrightarrow{*} bw'_2$$

$$aw'_1 \xleftrightarrow{*} avas' \longleftrightarrow buas' \xleftrightarrow{*} bw'_2$$

If we can decide if there exist s' such that $aw'_1 \leftrightarrow_R^* avas'$ and $bw'_1 \leftrightarrow_R buas'$, then we can decide the word problem.

When does $aw \leftrightarrow_R^* avas$ for some $s \in \{a, b\}^*$?

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- ▶ Otherwise, let p be the longest common prefix of aw and ava .
 - ▶ If $aw = paw'$, then $ava = pbv'$. Need to decide if $aw' \leftrightarrow_R^* bv's$. So need to decide if aw' can be written as $avas'$ for some s' . Do this recursively.

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 - ▶ If $aw = paw'$, then $ava = pbv'$. Need to decide if $aw' \leftrightarrow_R^* bv's$. So need to decide if aw' can be written as $avas'$ for some s' . Do this recursively.
 - ▶ If $aw = pbw'$, then $ava = pav'$. Need to decide if $bw' \leftrightarrow_R^* av's$. Decide if bw' can be written as $buas'$ analogously.

Prefix decomposition

Example

Let $R = \{baabbba \leftrightarrow aba\}$ and $w = bbaabbabaababaa$. Then w has prefix decomposition

$$|b|baabba|baab|\mathbf{aba}|a.$$

If a prefix decomposition contains a factor equal to a relation word, we say it has a *head* and otherwise that it is *headless*.

Adians algorithm

1. If the first letter of w is x or if the prefix decomposition of w is headless, then return w ;
2. If the head of the prefix decomposition of w equals u , then replace the head by v in w and go to step 1;
3. If the head of the prefix decomposition of w equals v , then replace the head by u in w and go to step 1.

Example

Let $R = \{baabbba \leftrightarrow aba\}$, $x = a$ and $w = bbaabbabaababaa$.

$|bbaabbabaababaa \rightarrow$

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Let $R = \{baabbba \leftrightarrow aba\}$, $x = a$ and $w = bbaabbabaababaa$.

$$\begin{aligned} &|b|baabba|baab|\mathbf{aba}|a \rightarrow \\ &\rightarrow |b|baabba|baab|baabbbaa \rightarrow \end{aligned}$$

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$$\begin{aligned} &|b|baabba|baab|\mathbf{aba}|a \rightarrow \\ \rightarrow &|b|baabba|\mathbf{baabbba}|bbaaa \rightarrow \end{aligned}$$

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Example

Let $R = \{baabbba \leftrightarrow aba\}$, $x = a$ and $w = bbaabbabaababaa$.

$|b|baabba|baab|**aba**|a \rightarrow$
 $\rightarrow |b|baabba|**baabbba**|bbaaa \rightarrow$
 $\rightarrow |b|**baabbba**|babbaaa \rightarrow$

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Example

Let $R = \{baabbba \leftrightarrow aba\}$, $x = a$ and $w = bbaabbabaababaa$.

$|b|baabba|baab|**aba**|a \rightarrow$
 $\rightarrow |b|baabba|**baabbba**|bbaaa \rightarrow$
 $\rightarrow |b|**baabbba**|babbaaa \rightarrow$
 $\rightarrow |ba|ba|ba|b|baaa|$

Example

Let $R = \{baabbbaa \leftrightarrow a\}$, $x = a$ and $w = bbbaa$.

$$\begin{aligned} &|b|baa|a| \rightarrow \\ \rightarrow &|b|baab|a|abbbaa \rightarrow \\ \rightarrow &|b|baabbbaa|bbbaabbbaa \rightarrow \\ \rightarrow &|ba|b|baa|a|bbbaa \rightarrow \dots \end{aligned}$$

Theorem (Nyberg-Brodda 2021, Theorem 4.3)

Let R be a 1-relation left cycle-free Thue system. If there is an algorithm which, given $x \in \{a, b\}$ and $w \in \{a, b\}^+$, decides if $\mathfrak{A}_R(x, w)$ terminates, then the word problem for R is decidable.

Corollary

If Adian's algorithm for R terminates on all inputs, then the word problem for R is decidable.

Example

Let $R = \{baabaa \leftrightarrow aba\}$, then $\mathcal{A}_R$ is given by

$$\begin{array}{lll}
 q_a a \rightarrow q_a q_a, & q_a b \rightarrow q_{ab}, & q_{ab} a \rightarrow q_{aba}, \\
 q_{ab} b \rightarrow q_{ab} q_b, & q_b a \rightarrow q_{ba}, & q_b b \rightarrow q_b q_b, \\
 q_{ba} a \rightarrow q_{baa}, & q_{ba} b \rightarrow q_{ba} q_b, & q_{baa} a \rightarrow q_{baa} q_a, \\
 q_{baa} b \rightarrow q_{baab}, & q_{baab} a \rightarrow q_{baaba}, & q_{baab} b \rightarrow q_{baab} q_b, \\
 q_{baaba} a \rightarrow q_{baabaa}, & q_{baaba} b \rightarrow q_{baaba} q_b, & \\
 q_{aba} \rightarrow baabaa, & q_{baabaa} \rightarrow aba. &
 \end{array}$$

Method	Count
Known mathematical results	222786
Knuth-Bendix	37624
Knuth-Bendix backtrack	502
Adian's SRS terminates (certified)	460
Adian's SRS terminates (no certificate)	135
Unsolved (Adian's SRS non-terminating)	75
Unsolved (Other)	50
Total	261632

Formal verification

For reproducibility, working with Florent Hivert (Université Paris Saclay), Assia Mahboubi (Nantes Université), Guillaume Melquiond (Université Paris Saclay) on generating formal proofs of decidability in the ROCQ (ex Coq) theorem prover.

Open problems

Problem 1

Is termination of Adian SRS decidable?

Problem 2

Is termination of Adian SRS a monoid invariant, i.e. preserved under change of generators?

Problem 3

Does there exist an instance of Adian's SRS for which the halting problem is undecidable? I.e. a fixed Adian SRS for which it is undecidable, given $w \in A^*$, whether all derivations beginning with w terminate.

Problem 4

Does every 1-relation monoid admit a complete SRS?

Quirky examples

Example

The 1-relation Thue systems for which the decidability of the word problem is unknown to us:

$$\{baabbbaba \leftrightarrow a\} \quad \{baaabaaa \leftrightarrow aba\}.$$

Example

The Adian SRS arising from the Thue system

$$\{baaababaa \leftrightarrow aba\},$$

admits looping derivations of length n for arbitrarily large n :

$$|ba|(|baa|)^n|baaa|aba| \rightarrow^* |baa|baa|ba|(|baa|)^n|baaa|aba|baabaa.$$

Example

The Adian SRS arising from the Thue system

$$\{baaababaabaa \leftrightarrow aba\},$$

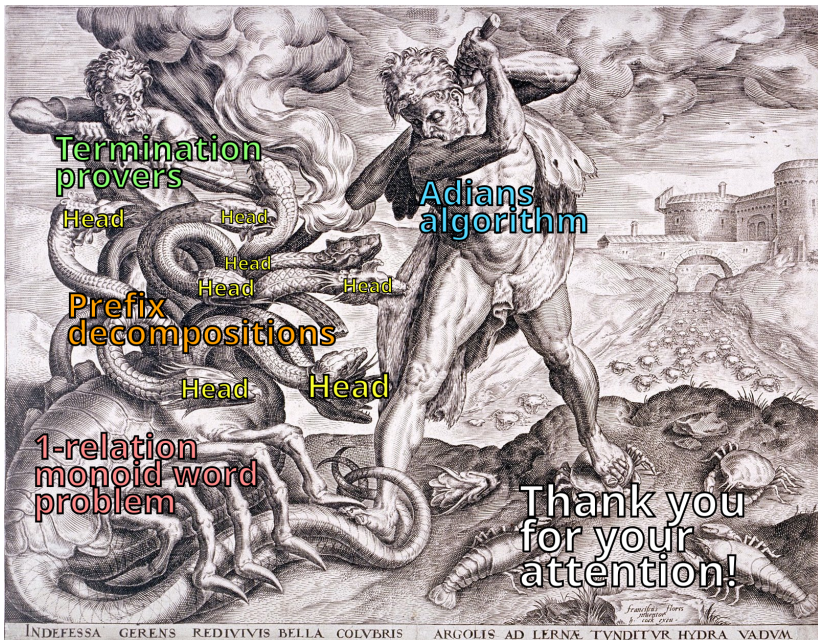
admits non-looping non-terminating derivation:

$$\begin{aligned} &|ba|(|baa|)^n|baaa|aba| \rightarrow^* \\ &\rightarrow^* |baa|baa|ba|(|baa|)^{4n+3}|baaa|aba|baabaabaabaa. \end{aligned}$$

Does it admit a looping derivation?

Adian SRS arising from these examples in ARI and SRS format, as well as a conversion tool are available at:

<https://github.com/reiniscirpons/AdianSRS>



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